Discovering adsorption rate laws from column data fails long before the rate constant is lost

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Abstract

Equation-discovery methods promise to learn the uptake rate law of an adsorption column from its data, but where they can is unknown. In a pre-registered computational study spanning four decades of Damköhler number (Da), three Péclet numbers, noise and isotherm error, we compared the Da at which the best discovery method stopped recovering the linear-driving-force law with the Da at which its rate constant stopped being practically identifiable. In all 37 of 96 conditions with both boundaries, discovery failed first, by 1.6 to 3.7 decades. With 5 % isotherm error no method reached 50 % success at any rate, while the rate constant remained identifiable from interior probes. Weak-form sparse regression worked at low Da and symbolic regression in a window near Da 1 to 10; neither reached the range where the rate constant stayed identifiable. Column data carry information about the rate long after its law can no longer be discovered.

A fixed-bed adsorption experiment reports concentrations and temperatures over time, most often at the column outlet. What an engineer needs from it is the uptake rate law, usually written as the linear driving force (LDF), $\partial q/\partial t = k\,(q^*(c,T) - q)$. Equation-discovery methods offer to find the law itself, not only its coefficient: sparse regression over a library of candidate terms [1, 2] and its weak-form [3, 4], ensemble [5] and exact best-subset [6] variants, symbolic regression [7, 8], and neural networks embedded in the governing equations [9]. For adsorption, one study has done this from column data: Santana et al. trained a neural uptake term on noisy outlet breakthrough curves and read the LDF form back out, with the isotherm given exactly and at one rate and one noise level [9]. Repeated database searches found no other study that discovers an adsorption rate law from column data. Where such a result holds, and where it stops holding, is not known.

Part of the answer is classical. Near local equilibrium the mass-transfer coefficient is poorly determined by column data [10]: finite rates then act only through an apparent dispersion [11], the LDF coefficient can be extracted wrongly from outlet breakthrough [12], and most of the output variability belongs to the isotherm [13, 14]; yet a lumped rate coefficient can be recovered from breakthrough in the constant-pattern regime [15] and estimated statistically from breakthrough curves [16, 17]. Batch studies warn that data near equilibrium cannot tell kinetic laws apart

[18, 19]. Equation discovery has its own limits. On a single transient that settles onto a slow manifold, as a breakthrough run does, the candidate terms become nearly collinear [20–22]; noise induces phase transitions in what can be recovered [23, 24]; unique recovery needs conditions non-chaotic dynamics generally lack [25]; and success maps of equation learning against sampling and noise exist for biological data [26]. Discovery has been coupled to identifiability by keeping only identifiable models [27, 28], and kinetic rate laws have been discovered in reaction engineering [29]. That discovering a law is harder than fitting its coefficient is therefore expected. How much harder, where, for which methods, and from which measurements has not, to our knowledge, been measured.

The question matters most where the law is not the textbook one. In the water-harvesting framework MOF-303 the water diffusion rate passes through a minimum at the steep step of the isotherm, and the LDF approximation is least accurate there [30, 31]; a loading-dependent rate has been fitted by hand [32]. A method that could discover such a law from column data would be useful. Whether one can, at the noise and isotherm accuracy of a real experiment, is what this study tests.

We asked, in a pre-registered computational study, for which regimes the adsorption rate law can be discovered from data a column experiment can produce, and whether discovery fails at the same boundary as the practical identifiability of the rate constant [33], or earlier. Ground truth came from a validated non-isothermal column model [34], for a Langmuir sorbent and a MOF-303-like step isotherm, over four decades of Damköhler number, three Péclet numbers, four noise levels and four levels of isotherm error, observed through interior probes of concentration, uptake and temperature, through probes without uptake, or at the outlet alone. Methods from each family above were run in the strongest configuration we could verify within declared cost limits, and identifiability was judged by profile likelihood on the same observations. The design, thresholds and verdict words were frozen before any method ran on the grid. Discovery failed first. In every slice of the design where both boundaries could be measured, the law stopped being discovered 1.6 to 3.7 decades of Damköhler number before its rate constant stopped being identifiable. With an isotherm error of 5 % no method reached 50 % success at any rate, and with 2 % only 8 of 24 slices reached it at any level, while k remained identifiable from interior probes and, where tested, from the outlet alone.

Results

Discovery fails long before identifiability

Both isotherms passed the positive control (lowest Da, no noise or isotherm error): the law was recovered, and k identifiable, in 135 of 135 replicates for each. (As first run, a search-range error withheld the step isotherm; the Langmuir verdict was the same; Supplementary Note 2, Extended Data Fig. 7.) As Da grows, the driving force $q^* - q$ that distinguishes the law shrinks below the noise, while the curves still constrain k (Fig. 2). H2b compares, in each slice of the design (one isotherm, one Péclet number, one noise level and one isotherm error), the Damköhler number at which the best discovery method falls to 50 % success, $\mathrm{Da}^*_{\mathrm{disc}}$, with the number at which k stops

being identifiable in half the replicates, $\mathrm{Da}^*_{\mathrm{ident}}$. Of the 96 slices, 37 had both boundaries on the grid. In all 37 the verdict was "*discovery fails before identifiability*", so H2b's pre-registered overall word is the same (Fig. 3, Table 1; every slice in Extended Data Figs. 1 and 2). The gap was large everywhere: from 1.6 to 3.7 decades of Da, with a median of 2.2 decades for the Langmuir sorbent (23 slices) and 2.5 for the step isotherm (14 slices); allowing for boundaries known only to lie between two levels, every gap is at least 1.35 decades (Methods). Discovery failed between Da = 0.11 and 13, while k stayed identifiable up to Da between 43 and 994. The best discovery method was exact best-subset selection in the weak form in 34 of the 37 slices and the weak-form ensemble in the other three, all without noise and with 0.5 % isotherm error.

Noise moved both boundaries, but not equally. With 0.5 % noise and an exact isotherm, discovery held to Da ≈ 4 to 7, and identifiability to Da ≈ 320 to 360 for the Langmuir sorbent and 560 to 740 for the step isotherm. At 2 % noise the discovery boundary fell to Da ≈ 1, and at 5 % to Da ≈ 0.1 to 0.3, near the bottom of the grid, while the identifiability boundary stayed between Da ≈ 57 and 172. A small error in the isotherm hurt discovery more than noise did: at 0.5 % noise, an isotherm error of 0.5 % moved $\mathrm{Da}^*_{\mathrm{disc}}$ from about 5 to about 0.1 to 0.2, more than a decade, while $\mathrm{Da}^*_{\mathrm{ident}}$ moved by less than 0.2 decade (Langmuir, from 320–360 to 300–320; step, from 560–740 to 690–870).

The 59 slices without a verdict are not missing data: in each, one probability had no 50 % crossing on the grid (pre-registered words "*no boundary on the grid for one of the two*"). In a descriptive check made after the verdicts, the discovery boundary was missing in all 59, and the identifiability boundary was on the grid in 56 of the 59. In 40, discovery never reached 50 % at any Da: all 24 slices with 5 % isotherm error, 15 with 2 % and one with 0.5 %. In 8, all without noise, it never fell below 50 %. In the other 11, success crossed 50 % more than once, which no single logistic crossing describes (Fig. 3), most clearly for symbolic regression (M7; next section).

The identifiability boundaries at the top of the grid need one caution, found in a second descriptive check made after the verdicts were seen. With exact data the O1 profile recovered k to within 5 % at every Da up to 100, but at Da = 316 and 1000 its median estimate was 0.75 to 0.91 and 0.39 to 0.54 times k: there the adsorption front crosses a probe in only a few of the 200 observation times. Identifiability there is limited by the time resolution of the data as well as by their information content. This does not affect H2b: in all 37 slices with a verdict, k was identifiable in at least 95 % of replicates at every Da ≤ 31.6, and the highest Da ≤ 100 at which it was identifiable in half the replicates lay at least 0.9 decades above the discovery boundary (Extended Data Fig. 3).

The verdict does not depend on how success is scored. When success also required the fitted k to lie within a factor of two of the truth (a pre-registered sensitivity analysis), the overall word and all 37 slice verdicts were unchanged, and two more slices, without noise and with 0.5 % isotherm error, gained a discovery boundary and the same verdict.

No single error ratio predicts the boundary, and errors-in-variables could not be tested

H2a asked whether one ratio, the realised driving force over the combined observation and isotherm error ($R = \delta/\varepsilon_{\mathrm{eff}}$), places the discovery boundary at the same R^* in every condition. Only one of twelve conditions crossed 50 % in R ($R^* = 123$, 95 % interval 113 to 135), so H2a has "*no verdict*" (Extended Data Fig. 8). In the others success was not monotone in R, peaking at 51 to 71 % on O1 and never exceeding 20 % at any level of R on O2.

H2c asked whether an errors-in-variables estimator (M5), which treats the measured isotherm as a regressor with error, moves the discovery boundary upward relative to the best strong-form comparator (18 slices at 2 % noise; Methods). No slice had a boundary for both, so H2c has "*no verdict*" at every isotherm error and is neither supported nor falsified. At this noise level the strong-form comparator succeeded in no replicate at any Da in 16 slices, while M5 reached at most 40 %; in the other two the comparator, symbolic regression, rose above 50 % and fell again.

Which methods discover the law, and where

Exact best-subset selection in the weak form (M4) was the most reliable method: its success crossed 50 % within the grid in 36 of the 96 slices, against 8 for weak-form sequential thresholding (M2) and 7 for the weak-form ensemble (M3) (Fig. 4c). No strong-form sparse method had a crossing in any slice. Symbolic regression (M7), run only at Pe ≈ 94 with noise of 0.5 or 2 % and isotherm error of 0 or 2 %, behaved differently from all the sparse methods. It failed at the lowest rates, where weak-form sparse regression worked best, and succeeded at intermediate rates: up to Da ≈ 32 without isotherm error, and between Da $= 1$ and 10 with 2 % isotherm error, where no sparse method reached 50 % (Fig. 4a,b; Extended Data Fig. 5). The pre-registered rule did not select symbolic regression as the best method in the slices it shared with sparse regression, because a falling logistic curve cannot describe success that is zero at the lowest rates. Taking its last level with at least 50 % success instead would not change H2b: in the four such slices with a verdict, that level lay 1.1 to 1.4 decades below the identifiability boundary. The two families therefore cover different ranges of Da, and neither covers the range in which k remains identifiable. declared demonstration at one noise level, recovered the law in one of its nine fits, at Da $= 3.16$. Slow-manifold discovery (M9), which targets the isotherm rather than the rate law, recovered its own target only at the highest rate and without noise (67 % of replicates at Da $= 1000$, pooled over isotherms and Péclet numbers), and in at most 17 % elsewhere.

Where open discovery fails, the law can still be chosen among a few candidates. In a post hoc check (E6), the LDF law, the quadratic driving force and the pseudo-second-order law, the three most used in adsorption practice, were each fitted by the likelihood used for identifiability and the best fit chosen (Pe ≈ 94, noise 0.5 and 2 %, isotherm error 0 and 2 %). Choice carried further than open discovery: the LDF law was chosen in all 45 replicates at every Da ≤ 10 with an exact isotherm, and in at least 78 % with 2 % isotherm error, where sparse regression failed at every rate (Fig. 4d). It did not reach the identifiability boundary: with an exact isotherm, the LDF law was

chosen in only 31 of 180 replicates at Da = 31.6 and in none from Da = 100 on, at both noise levels alike, so the failure is systematic, not statistical. Sampling explains part of it. With twice as many observation times (E8, post hoc; exact isotherm), every boundary moved to higher Da: discovery by 0.4 to 0.7 decade, choice among the laws by 0.2 to 0.5 decade, identifiability by up to 0.3 decade or beyond the grid. Yet no replicate chose the LDF law at Da = 100, and discovery still failed at least 1.2 decades before identifiability.

Without probes of uptake, k stays identifiable but the law is not discovered

Real columns are rarely fitted with probes inside the bed. With concentration and temperature probes but no uptake (O2), uptake must first be reconstructed from the gas-phase mass balance. Discovery failed on O2 everywhere: no slice had a 50 % boundary, and no method succeeded in more than 29 % of the replicates at any level of any slice. Identifiability, computed on the subgrid where every point of the profile is a full column solve (Pe ≈ 94, noise 0.5 and 2 %, isotherm error 0 and 2 %), held to Da ≈ 57 to 286 from probes without uptake, and to Da ≈ 38 to 172 from the outlet alone (O3). These profiles are costly, so they used three replicates per level, and their boundaries are correspondingly coarse. In the same slices, the last level at which any method discovered the law from probes of c, q and T, a far richer observation, lay at Da = 3 to 32, so the outlet curve still determined k 0.25 to 1.5 decades beyond it.

A loading-dependent rate is found almost only at one rate

For the step isotherm the study also simulated a rate that falls to 20 % of its value across the isotherm step (L2), a form motivated by the measurements in MOF-303; this part is exploratory. The state dependence was discovered, by the pre-registered rule, almost only at Da = 1, where the best method succeeded in 77 to 100 % of all 60 replicates at every noise level and isotherm error, against at most 58 % at any other level and none at Da ≤ 0.32 without noise (Extended Data Fig. 6; best over methods and Péclet numbers, so an upper bound for any single method). The loading did pass the step at low Da, reaching 41 to 47 % of capacity against a step midpoint at 36 %. A likely reason lies in the test itself: the library holds no term that can represent a rate falling sigmoidally across the step, so the rule asks whether the selected polynomial terms, which must include q and q^*_{meas}, imply a rate falling by at least 40 % across the observed loading, and success then follows which terms are selected.

Discussion

The rate law of a fixed-bed column could be discovered only over a narrow part of the range in which its rate constant could be measured. In every slice of the design where both boundaries existed, discovery failed between 1.6 and 3.7 decades of Damköhler number before identifiability did. For practice this sets an order of operations. Where the isotherm is known almost exactly (an error of 0.5 % with 0.5 % noise already confined discovery to Da ≤ 0.2) and the uptake is slow compared

with the passage of the front (Da of order one or below), sparse regression in the weak form can propose the law. Up to Da ≈ 10 the law can still be chosen among a few candidates by likelihood, in our post hoc test even with 2 % isotherm error. Beyond that it has to be assumed, and the data are then still good for its coefficient, from interior probes up to one and a half to nearly four decades of Da beyond the discovery boundary, and, in the few conditions tested, somewhat beyond the discovery range from the outlet alone.

Why is discovery so much more fragile? The two tasks ask different things of the same data. Identifiability asks whether a known law with one free coefficient fits well only near the true value. Discovery asks the data to choose that law among many, and, in the success rule used here, to return the coefficients of q^*_{meas} and q in the ratio -1 within 10 %. An isotherm error enters discovery as an error in a regressor, which biases exactly that ratio. In the identifiability fit the same error is absorbed into k, and the estimate stays within a factor of two of the truth. The observation that an isotherm error of 0.5 % moved the discovery boundary by more than a decade, while it moved the identifiability boundary by less than 0.2 decade, is consistent with this reading, although the study was not designed to separate the mechanisms. The asymmetry of the two success rules does not make the gap: in a post hoc check (E4), dropping the ratio test from discovery, requiring the interval to lie within 10 % of k, or judging identifiability by the point estimate alone kept every verdict, with gaps of 0.9 to 3.8 decades. A second effect is classical: as Da grows, uptake approaches equilibrium and the law's two terms become nearly collinear on the slow manifold [21, 22]. In a descriptive check made after the verdicts, the smaller singular value of the standardised pair $\{q, q^*_{\mathrm{meas}}\}$ fell from about 0.75 of the larger at Da $= 0.1$ to about 0.001 at Da $= 1000$. Exact data still separate the two terms, which is why discovery never failed without noise, but noise and isotherm error soon exceed the shrinking direction that distinguishes them.

Where do real columns sit on this map? Rate constants and bed sizes reported for water on MOF-303 [30] and for carbon dioxide on zeolite 13X [35] place laboratory packed beds at roughly Da $= 10^2$ to 10^4, and thin coated layers possibly lower; these are order-of-magnitude estimates from a few sources, some read from figures. Practice therefore lies mostly beyond every discovery boundary measured here: at its lower end the data would still give k, and at its upper end, beyond the identifiability boundaries found here, neither the law nor its coefficient.

The closest prior work recovered the LDF law from noisy outlet curves at one rate, with the isotherm given exactly [9]. Our results place that success at one end of a wider map: with an exact isotherm, discovery from probes survives 2 to 5 % noise at low Da, but outside that corner it fails. For water in MOF-303, where the rate depends on loading at the isotherm step [30, 31], the discoverability of the law remains open: our library could not represent a sigmoidal $k(q)$, so the H2d test measured the selection of polynomial terms more than that.

The study has limits that bound these conclusions. All data were simulated by one validated column model with the LDF law as the truth, so the boundaries hold for that model, the two isotherms and the error models used; real columns add errors in geometry, dispersion and heat transfer that were held exact here. Observation through interior probes of uptake (O1) is a ceiling

that experiments rarely reach. The identifiability criterion asks whether the 95 % profile interval lies within a factor of two of k; it is a criterion of precision and accuracy, not of calibration. In an exploratory check the intervals contained the true k in only about 10 % of O1 rows (23 % on O3; Extended Data Fig. 4), because they are narrower than the small biases from time resolution and from errors in the inputs to q^*_{meas}. At the two highest Da levels the time resolution also biased the estimate itself, downwards by up to a factor of 2.5, so identifiability there is partly a property of the sampling; H2b does not depend on those levels. Symbolic regression ran on a subgrid, the neural pipeline only as a demonstration, and a Kolmogorov–Arnold network only in an exploratory run on its declared subgrid, where it recovered the law in none of 216 fits; each limit was set by cost rules declared before the grid ran. Combining method families was not tested, nor was errors-in-variables fitting in the weak form. Finally, five errors in our own code and one solver setting were found and corrected during the study. The one that mattered most, a wrong coefficient in the exact update of the O1 likelihood found in a later review of the code, moved half of the identifiability boundaries to lower Da and raised the number of slices with a verdict from 30 to 37. All earlier results are public (Supplementary Note 2); no correction changed a pre-registered verdict word.

Three directions follow. Discovery would gain most from better isotherms: the boundaries moved more with isotherm error than with noise, and in a post hoc test, shifting the step of the MOF-303-like isotherm by 1 % relative humidity kept discovery at or below 31 % at every rate. An isotherm whose step is placed to better than 1 % relative humidity is worth more to discovery than quieter sensors. Experiments can also be designed to discriminate among candidate laws, choosing probe positions and sampling rates that widen the range where the LDF law is distinguishable from its alternatives. And the map should be tested on experimental breakthrough data with interior probes at moderate Da, where it predicts that discovery first becomes possible.

Methods

Ground truth

Ground truth came from the finite-difference reference solver of the companion study [34]: a one-dimensional, axially dispersed, non-isothermal model of a packed column, verified there against closed-form solutions. Uptake followed one of two laws,

$$\text{L1:}\quad \frac{\partial q}{\partial t} = k\,(q^*(c,T) - q), \tag{1}$$

$$\text{L2:}\quad \frac{\partial q}{\partial t} = k(q)\,(q^*(c,T) - q), \qquad k(q) = k_0\Big[1 - 0.8\,s\Big(\frac{q/q_{\max} - \theta}{0.05}\Big)\Big], \tag{2}$$

where s is the logistic function and θ is the fractional loading at the isotherm's step midpoint, so that under L2 the rate falls to 20 % of k_0 across the step. Two adsorbents were simulated with the same column geometry and flow (Supplementary Table 1): the companion study's favourable Langmuir sorbent, and water on a MOF-303-like framework, a cooperative (Type V) isotherm with its step at $P/P_0 = 0.15$, fed at 30 % relative humidity and 30 °C. The Damköhler number $\mathrm{Da} = k\,t_{\mathrm{stoich}}$

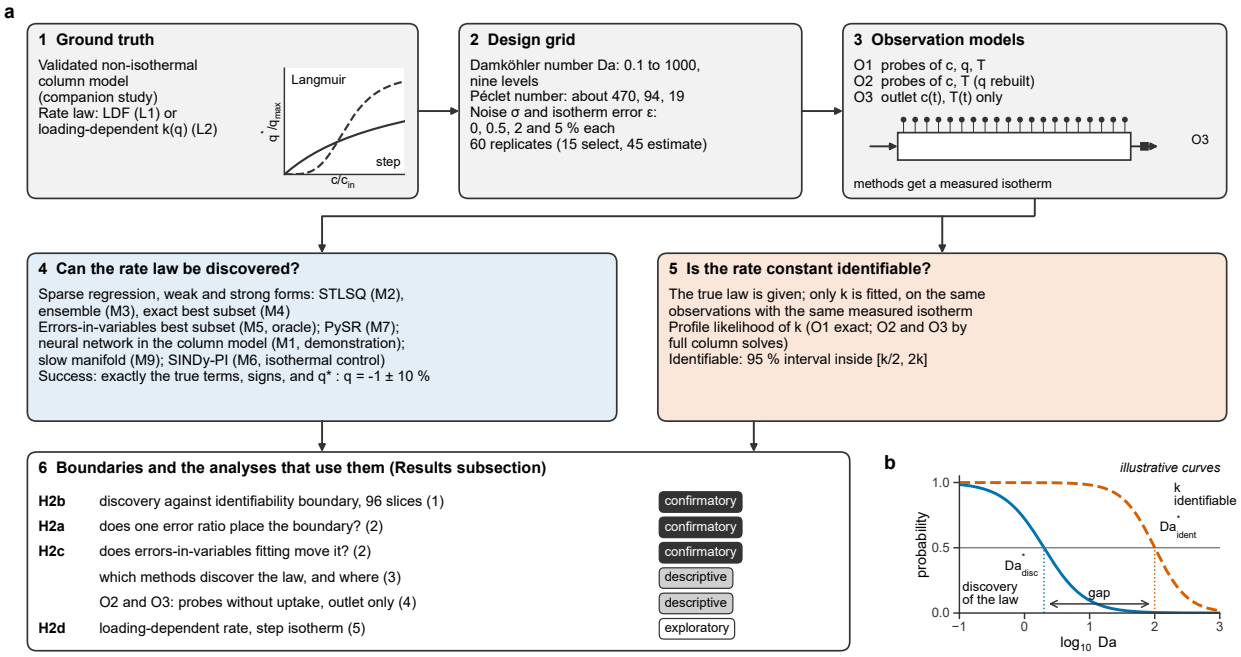


Figure 1: **Study design.** **a**, Breakthrough data are simulated by a validated column model for a Langmuir sorbent and a MOF-303-like step isotherm (box 1; isotherm shapes drawn from the model), over the design grid (box 2), and observed through interior probes with uptake (O1), probes without uptake (O2) or the outlet alone (O3) (box 3). The same observations, with the same measured isotherm, go to the discovery methods (box 4) and to a profile-likelihood test of k with the true law given (box 5). Box 6 lists the analyses with the Results subsection that reports each and whether it was confirmatory, exploratory or descriptive. **b**, What H2b compares in each slice: the Damköhler number at which the best discovery method falls to 50 % success ($\mathrm{Da}^*_{\mathrm{disc}}$) and the one at which k stops being identifiable in half the replicates ($\mathrm{Da}^*_{\mathrm{ident}}$). The curves in **b** are illustrative.

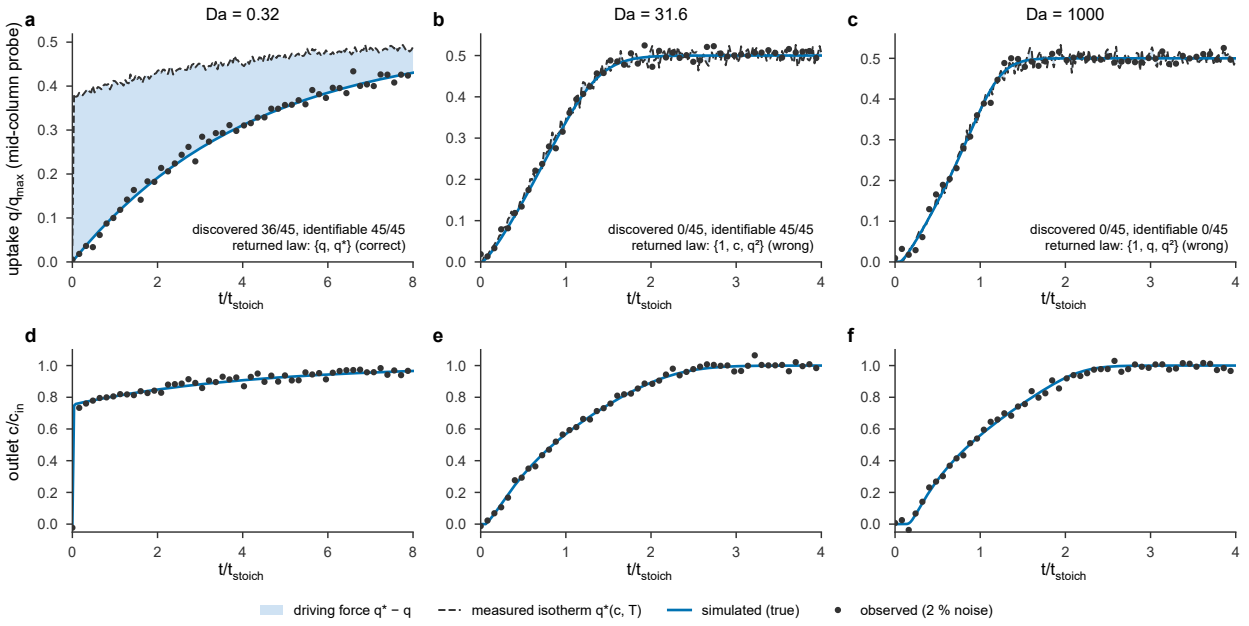


Figure 2: **What the data look like where discovery works, where only identifiability works, and where neither does.** Langmuir sorbent, Pe ≈ 94, 2 % noise and an exact isotherm, at Da $= 0.32$, 31.6 and 1000 (columns). **a**–**c**, A mid-column probe as the methods see it: observed uptake (points, every fourth time shown), simulated uptake (line) and the measured isotherm evaluated on the observed c and T (dashed); the shaded band is the driving force $q^* - q$, which shrinks below the noise as Da grows. Each panel gives the number of the 45 estimation replicates in which the law was discovered and k was identifiable, and the law the best method returned for one replicate. **d**–**f**, The outlet breakthrough curve of the same replicate.

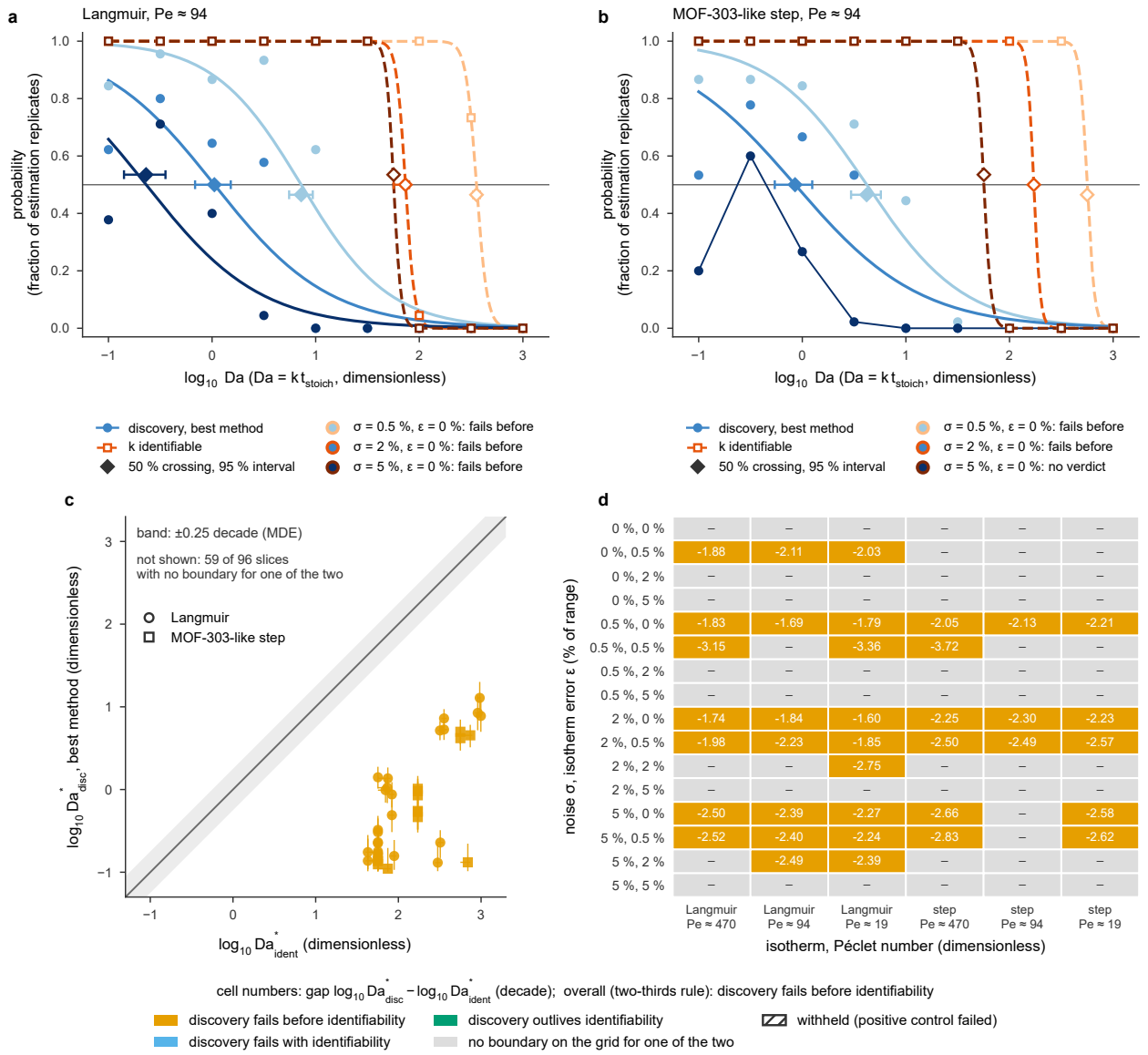


Figure 3: **Discovery fails long before identifiability.** **a**,**b**, Fraction of the 45 estimation replicates in which the best discovery method recovered the LDF law (filled circles, blue) and in which k was practically identifiable (open squares, orange), against Da, for an exact isotherm and noise of 0.5, 2 and 5 % (light to dark), at Pe ≈ 94; **a**, Langmuir sorbent; **b**, MOF-303-like step isotherm. Curves are the logistic fits that define the boundaries, drawn only where a 50 % crossing exists; otherwise the measured fractions are joined. Diamonds mark the crossings with 95 % bootstrap intervals. **c**, $\mathrm{Da}^*_{\mathrm{disc}}$ against $\mathrm{Da}^*_{\mathrm{ident}}$ for the 37 slices with both boundaries; the grey band is the pre-registered verdict threshold, the minimum detectable effect (±0.25 decade), around equality. **d**, The H2b verdict of every slice (rows: noise and isotherm error; columns: isotherm and Péclet number) with the gap $\log_{10}\mathrm{Da}^*_{\mathrm{disc}} - \log_{10}\mathrm{Da}^*_{\mathrm{ident}}$ in decades. Grey cells have no boundary on the grid for one of the two.

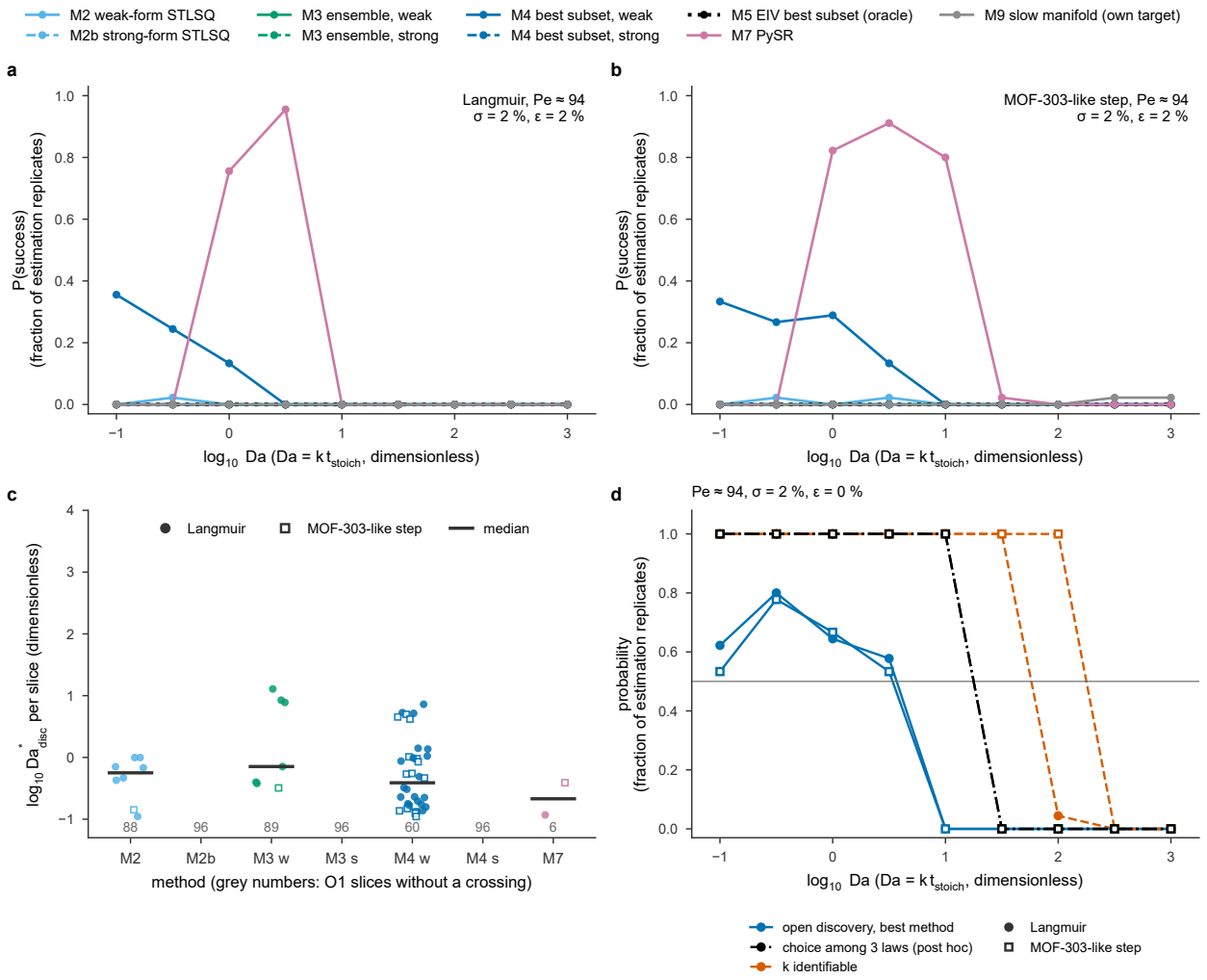


Figure 4: **Methods compared.** **a**,**b**, Success of every method against Da at 2 % noise and 2 % isotherm error, Pe ≈ 94; **a**, Langmuir; **b**, step isotherm. Symbolic regression (M7, orange) succeeds only in a window of intermediate Da, where every sparse method stays below 50 %. **c**, $\log_{10} \mathrm{Da}^{*}_{\mathrm{disc}}$ of each method in every slice where it crosses 50 %; the grey numbers count the slices without a crossing. **d**, Choosing among three candidate laws (LDF, quadratic driving force, pseudo-second order) by likelihood (E6, exploratory, declared after the verdicts; black, dash-dotted), beside open discovery by the best method (blue) and identifiability of k (orange, dashed), at Pe ≈ 94, 2 % noise and an exact isotherm; circles, Langmuir; open squares, step isotherm. Fraction of the 45 estimation replicates in which the LDF law was chosen, recovered or identifiable.

Table 1: **Pre-registered hypotheses and their verdicts.** Words in quotation marks are the pre-registered verdict words. Intervals are 95 % bootstrap intervals over the estimation replicates.

	Question	Result	Verdict
Positive control	Do discovery and identifiability both succeed at the lowest Da, without noise or isotherm error ($\geq 90\,\%$)?	Langmuir 135/135 and 135/135; step 135/135 and 135/135	Both isotherms enter every verdict
H2b	Does discovery fail before, with, or after identifiability (± 0.25 decade)?	37 of 96 slices with both boundaries; gap -1.60 to -3.72 decades (median -2.23 Langmuir, -2.50 step)	"*discovery fails before identifiability*" (37 of 37 slices)
H2b, k-aware	As H2b, with success also requiring $\hat{k}$ within a factor of 2	39 of 96 slices with both boundaries; the 37 verdicts unchanged	"*discovery fails before identifiability*" (39 of 39)
H2b, any method (exploratory)	How small a gap does any method allow?	Each slice's best method judged on the estimation replicates gives the same 37 gaps; symbolic regression (four slices, success in a window of Da) last reached 50 % 1.1 to 1.4 decades below $\mathrm{Da}^{*}_{\mathrm{ident}}$	Descriptive
H2a	Is the boundary at one value of $R = \delta/\varepsilon_{\mathrm{eff}}$ (spread < 0.5 decade)?	One of 12 conditions crosses ($R^{*} = 123$, 113 to 135)	"*no verdict: fewer than two conditions have a crossing*"
H2c	Does errors-in-variables fitting move the boundary up by ≥ 0.25 decade (two thirds of slices) at $\varepsilon > 0$?	No slice with a boundary for both estimators	"*no verdict*" at every ε
H2d (exploratory)	Is a loading-dependent rate discovered for the step isotherm?	At $\mathrm{Da} = 1$ in 77 to 100 % of replicates (best method); at most 58 % elsewhere	Descriptive

counts kinetic time constants per passage of the adsorption front, with t_{stoich} the stoichiometric time of the column (7200 s for Langmuir and 31 050 s for the step isotherm). Nine levels from 10^{-1} to 10^{3} at half-decade spacing therefore span k from 1.39×10^{-5} to $0.139\,\mathrm{s}^{-1}$ for Langmuir and from 3.22×10^{-6} to $0.0322\,\mathrm{s}^{-1}$ for the step isotherm. Three axial-dispersion coefficients, $D_L \times 0.2$, $\times 1$ and $\times 5$ of the default, completed the grid. They give column Péclet numbers $\mathrm{Pe} = uL/D_L$ of about 470, 94 and 19, and the levels are named by these values below (the code and the pre-registration label them by the D_L multiplier, so its "$\times 0.2$" level is the highest Pe). The grid has 81 solves at 400 cells and 400 stored times, each run until the outlet reached 95 % breakthrough, plus 9 isothermal solves for a control described below. The manifest that records every parameter is public.

Observation models, noise and isotherm error

Every method and every identifiability calculation read the data through one function, so all saw the same corrupted observations. Three observation models were used: interior probes of c, q and T (O1), interior probes of c and T with q hidden (O2), and the outlet $c(t)$ and $T(t)$ only (O3). O1 and O2 used 20 equally spaced probes; all models used 200 times spread evenly over the solved horizon. Noise was independent Gaussian with standard deviation $\sigma \in \{0, 0.5, 2, 5\}$ % of each channel's range. Methods were never given the true isotherm. They were given a measured one, $q^{*}_{\mathrm{meas}}(c,T) = q^{*}(c,T)\,[1 + \varepsilon\, g(c/c_{\mathrm{in}})]$ with $\varepsilon \in \{0, 0.5, 2, 5\}$ %, where g is a smooth random function of four Fourier modes on $[0,1]$, scaled to unit root-mean-square and drawn once per replicate. Every random draw was seeded by a hash: the noise by (cell, replicate, σ, observation model) and the isotherm error by (cell, replicate), so each observation can be regenerated on its own and every observation model sees the same measured isotherm. A replicate, one such observation of one cell, was the unit of analysis. Discovery on O1 and O2 and identifiability on O1 used 60 replicates per cell; the costlier O2 and O3 identifiability profiles used 3, and the M1 demonstration 1.

Discovery methods

Each method ran in the strongest configuration we could verify on known-answer systems before the pre-registration was frozen, and a method that could not be run that way was reported as such rather than replaced by a weaker one. The candidate library was $\{1, c, q, T, cq, q^2, c^2, q^{*}_{\mathrm{meas}}\}$; under L1 the true law is $k(q^{*}_{\mathrm{meas}} - q)$. The methods were sparse regression on a Savitzky–Golay estimate of $\mathrm{d}q/\mathrm{d}t$ (strong form) and on its weak, integral form with compactly supported test functions [3, 36], each with three solvers: sequentially thresholded least squares [1], a bagged ensemble with an inclusion threshold of 0.6 [5], and exact best-subset selection by enumeration (M4; exact subset selection as in [6], here scored by the Bayesian information criterion); errors-in-variables best-subset regression given the true error levels (M5, an oracle); implicit sparse regression without any isotherm on an isothermal control (M6, SINDy-PI [37]); symbolic regression (M7, PySR [7]); a universal differential equation with a neural network in the uptake term, trained through the column model and read out by sparse and symbolic regression (M1, the pipeline of [9]); and regression of the state on q-free features, which returns the isotherm where uptake is at equilibrium (M9 [22]). On O2, uptake was

first recovered from the gas-phase mass balance, and the three sparse solvers were then run on it in both forms; M5, M7 and M9 were not run on O2, as the pre-registered applicability table states. 2 h per fit on the machine available, so it ran as a declared demonstration: Langmuir, Pe ≈ 94, $\sigma = 2\,\%$, $\varepsilon = 0$, O3, the nine Da levels, one replicate. The best parameters seen during training were kept. As first run, a column solve that exhausted its step budget during training ended the fit with no readout (four of the nine fits); the corrected code keeps the best parameters in that case, and both runs are reported (Supplementary Note 2). A Kolmogorov–Arnold network [38] failed its known-answer test and its cost gate before the freeze and was not run on the grid. Method settings, solver thresholds and their known-answer tests are listed in Supplementary Note 1.

Success

Discovery succeeded when the selected terms were exactly those of the true law, every sign was correct and, for L1, the ratio of the q^*_{meas} and q coefficients was -1 within 10 %. The value of k itself was not required. A method that raised an error counted as a failure; no replicate, cell or output was excluded.

Identifiability

Practical identifiability of k was judged by profile likelihood [33] with the true law known, on the same observations and with the same measured isotherm as the discovery methods: k was practically identifiable when the 95 % profile interval (likelihood-ratio threshold 3.84) lay within $[k/2, 2k]$. On O1 the likelihood summed over all probes the fit of the known law, integrated exactly for piecewise-linear $q^*(t)$ along the observed c and T. On O2 and O3 every point of the profile was a full column solve. The O1 likelihood involves only the local uptake law along the observed c and T, so errors in dispersion, heat transfer or geometry do not enter it; on O2 and O3 the column model is taken as exact, as it is in the simulations. The profile searched a coarse logarithmic grid of k and refined it around the interval (41 points per pass on O1, 21 on O2 and O3, as pre-registered). Three corrections to the identifiability profiles were made during the study and are reported in full (Supplementary Note 2). First, on O2 and O3 a measured isotherm with $\varepsilon > 0$ could stall the stiff solver at large trial k for days; each solve was therefore capped at 600 s and a capped solve was scored as a failed solve (amendment A5; two O3 rows had capped solves, none next to its interval). Second, the O1 grid first spanned 10^{-5} to $10\,\mathrm{s}^{-1}$, which excluded the step cells at Da = 0.1; it was widened to 10^{-7} to $10\,\mathrm{s}^{-1}$, a peak on the grid edge was made an error, and all O1 rows were recomputed (amendment A6). Before that recomputation, the O1 refinement was found to use 21 points per pass instead of the pre-registered 41 and was set to 41, so every corrected O1 row uses the declared settings. Third, a later review of the code found that the O1 update, meant to be exact, carried a wrong coefficient: it differed from the exact solution only by a term of order $(kh)^2$ (h the interval between observations), so a check made at small kh had passed, but it misfit the data when kh was of order one or larger, that is, at high Da. The update was corrected and tested against a numerical ODE solver for kh from 10^{-4} to 100, and all 51 840 O1 rows were recomputed

with the analysis unchanged (amendment A7 records this correction and the refinement setting).

Boundaries and verdicts

For each slice the boundary Da^* was the 50 % crossing of an unpenalised logistic regression of replicate outcomes on $\log_{10}\mathrm{Da}$, with a percentile bootstrap interval (2000 resamples within each Da level). No boundary was reported, and no verdict issued, when the fitted curve did not cross 50 % inside the tested range or rose with Da instead of falling; the range was reported instead. Before the freeze the estimator's coverage on planted boundaries was 0.94 for a nominal 0.95, with a bias of +0.012 decade. The best discovery method of each slice was chosen on replicates 0–14 and its boundary estimated on replicates 15–59, which removes the inflation that choosing and estimating on the same replicates causes (0.07 decade in simulation); the identifiability boundary was estimated on the same 45 replicates. The best method was the one with the largest crossing on the selection replicates, or, if none crossed, the one with the highest mean success. Only rate-law methods that ran on every cell of the slice competed: M5, which knows the true error levels, and M9, which targets the isotherm, were excluded.

Three hypotheses were confirmatory, each with verdict words fixed in advance. H2a: success collapses onto one curve in $R = \delta/\varepsilon_{\mathrm{eff}}$, the driving force relative to the combined error, with the spread of R^* across conditions under half a decade. H2b: on O1, for each slice (σ, ε, isotherm, Pe; 96 in all), the gap $\log_{10}\mathrm{Da}^*_{\mathrm{disc}} - \log_{10}\mathrm{Da}^*_{\mathrm{ident}}$ was called "*discovery fails before identifiability*" at -0.25 decade or less, "*discovery outlives identifiability*" at $+0.25$ or more, and "*discovery fails with identifiability*" in between; the hypothesis-level verdict was the word held by at least two thirds of the slices with a verdict. Slices share ground-truth solves and isotherm-error draws, so they are not independent, and the two-thirds rule summarises them without a formal error rate. H2c: errors-in-variables regression moves the discovery boundary upwards by at least 0.25 decade in at least two thirds of the slices with a verdict at a given $\varepsilon > 0$, tested at $\sigma = 2\,\%$ on O1 in the strong form against the best strong-form method that is not errors-in-variables, in 18 slices (three ε, two isotherms, three Péclet numbers); a slice has a verdict only if both estimators have a boundary on the grid, and the refined grid placed around those boundaries is solved only if they exist. With 45 estimation replicates the H2b rule returned a wrong directional verdict with probability 0.001 when there was no gap, and the correct one with probability 0.965 at a gap of 0.40 decade, the smallest effect of interest. A positive control guarded every verdict: in each isotherm's easiest cells (lowest Da, $\sigma = \varepsilon = 0$, O1, the three Péclet numbers pooled) the best method had to succeed and k had to be identifiable in at least 90 % of the estimation replicates, or that isotherm's verdicts were withheld. O2 and O3 were reported descriptively, and H2d, whether the loading-dependent rate of L2 can be discovered, was exploratory.

Pre-registration and amendments

The design was frozen on 11 October 2026 (commit `7dcc2fe`) before any discovery method ran on any grid cell, and every confirmatory driver refused to run unless the pre-registration matched

its latest freeze commit. (The study's history was moved from the companion project's repository into its own; the pre-registration's amendment headers cite the original commit identifiers, and a published table maps them to the ones given here.) Seven dated amendments are appended to it. A1 to A4 were made before the grid ran: 60 replicates split into selection and estimation sets, the smallest effect of interest, the oracle kept out of the pooled verdicts, Pe as a slicing axis, errors counted as failures, the decision to run the grid before peer review, and the M1 demonstration. A5 (the solve cap) was made while the grid ran, and A6 (the O1 grid range) and A7 (two corrections to the O1 code, recorded without changing the declared method) after the verdicts were first computed. The verdicts as first run, before the last correction and corrected are all public, and the paper reports them (Supplementary Note 2).

Exploratory analyses

Every exploratory analysis was declared in writing, with what it would compute, before it ran, and none enters a confirmatory test. Three were declared before the verdicts: M1 trained at the scale a universal differential equation needs (E1), the Kolmogorov–Arnold network on its declared subgrid (E2), and how often an interval that meets $[k/2, 2k]$ misses the true k (E3). E1 needs a graphics processor and has not run; it is not reported here. Five more were declared after the verdicts, prompted by a review of the draft, and are labelled post hoc: H2b with matched success criteria for the two tasks (E4), the conditioning of the library against Da (E5), choice among three candidate laws by likelihood (E6), a systematic shift of the step of the measured isotherm (E7), and the sampling rate, 200 against 400 observation times (E8, declared after E6's results). E6 fits each law by implicit Euler along the same upsampled path of q^*_{meas} as the identifiability profile, with its rate constant profiled on a logarithmic grid and refined by a bounded one-dimensional search; all three laws have one parameter, so the Bayesian information criterion picks the largest likelihood. In E8 every method runs unchanged, so its windows, defined in samples, span half the time at 400 samples. Three descriptive checks were also made after the verdicts and are labelled so: the accuracy of the O1 profile with exact data, H2b without the two highest Da levels, and a classification of the slices without a verdict. All are reported in Supplementary Note 3.

Boundaries with perfect separation

Where every replicate is identifiable at one level of Da and none at the next, the unpenalised logistic fit has no finite maximum-likelihood slope: the optimiser stops at a steep curve whose crossing lies between the two levels, near their geometric midpoint. Every bootstrap resample reproduces the same data, so the interval has zero width. Such a boundary is reported as lying between those two levels; the gap used by H2b is computed from it exactly as from any other crossing. Such a boundary is interval-censored. This was so for the identifiability boundary in 18 of the 37 slices with a verdict; placing each of these at the lower of its two levels, nearest the discovery boundary, every gap is still at least 1.35 decades, so no verdict depends on where in its interval the boundary lies, and the gaps quoted in the main text are point estimates within these bounds.

Software and computation

The grid ran on a single computer without a graphics processor, one job at a time, from 12 October to 1 November 2026; the corrected O1 identifiability rows and the four M1 refits were computed from 1 to 4 November. After the last correction, the O1 identifiability rows were recomputed on 8 November on cloud CPU sessions (Kaggle) with the same code, and a random sample of rows recomputed locally agreed to a relative 10^{-6}; the exploratory model-selection and sampling analyses ran the same way. Package versions are pinned in the repository.

Use of AI tools

During the preparation of this work the author used a generative AI assistant (Claude, Anthropic) as a tool to assist with drafting and language editing, with writing and checking code, and with checking bibliographic records. The author designed the study, made every scientific decision, reviewed and edited all content, and takes full responsibility for the content of the published article.

Data availability

The ground-truth manifest, every results file, the run log and the as-run and corrected verdicts are public in the study repository and archived on Zenodo (doi:10.5281/zenodo.23325994). The ground-truth fields are regenerated from the manifest by `p2_generate.py`.

Code availability

All code is public at https://github.com/abdulhamidbatayhi123/ai-kinetics-discoverability under the MIT licence and archived with the data. The solver is imported unchanged from the companion study's archive (doi:10.5281/zenodo.23323685).

Author contributions

A.B. is the sole author and conceived and designed the study. The use of AI tools is described in Methods.

Competing interests

The author declares no competing interests.

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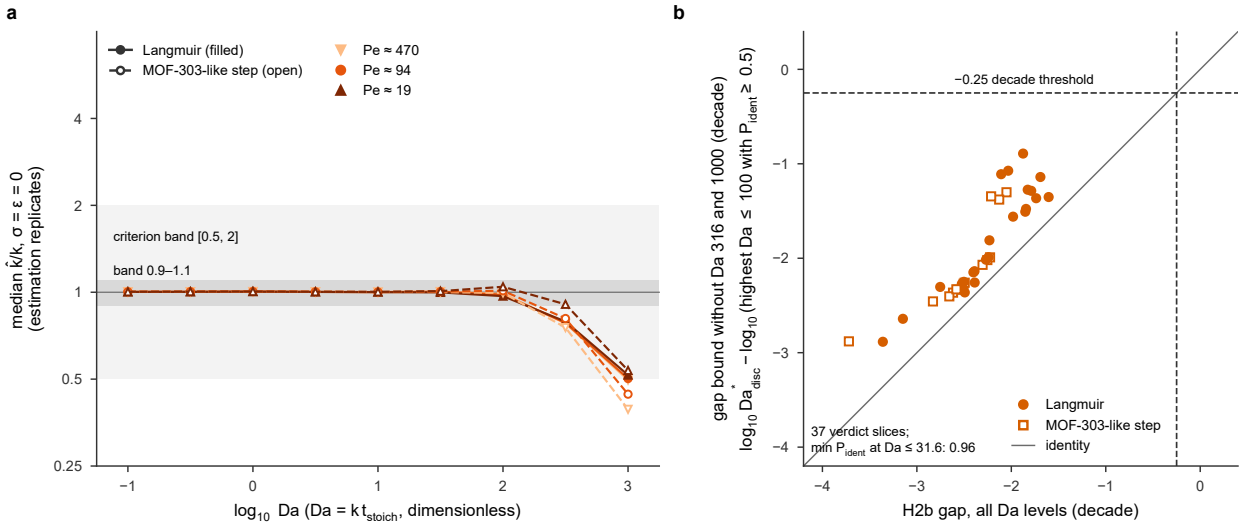
Extended Data



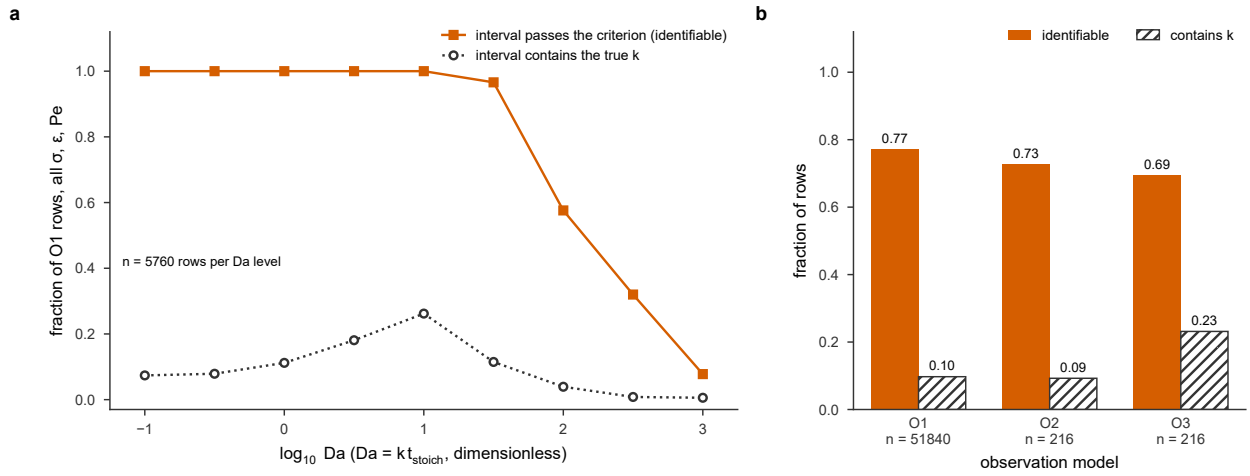
Extended Data Fig. 1: **Every H2b slice, Langmuir sorbent.** Fraction of the 45 estimation replicates in which the best method recovered the law (blue circles) and in which k was identifiable (orange open squares), against $\log_{10}$ Da; rows, noise σ and isotherm error ε; columns, Péclet number. Smooth curves are the logistic fits that define the boundaries, drawn only where a 50 % crossing exists; otherwise the measured fractions are joined. The square beside each panel gives the verdict (orange, "*discovery fails before identifiability*"; grey, no boundary on the grid for one of the two) and the number gives the gap in decades.



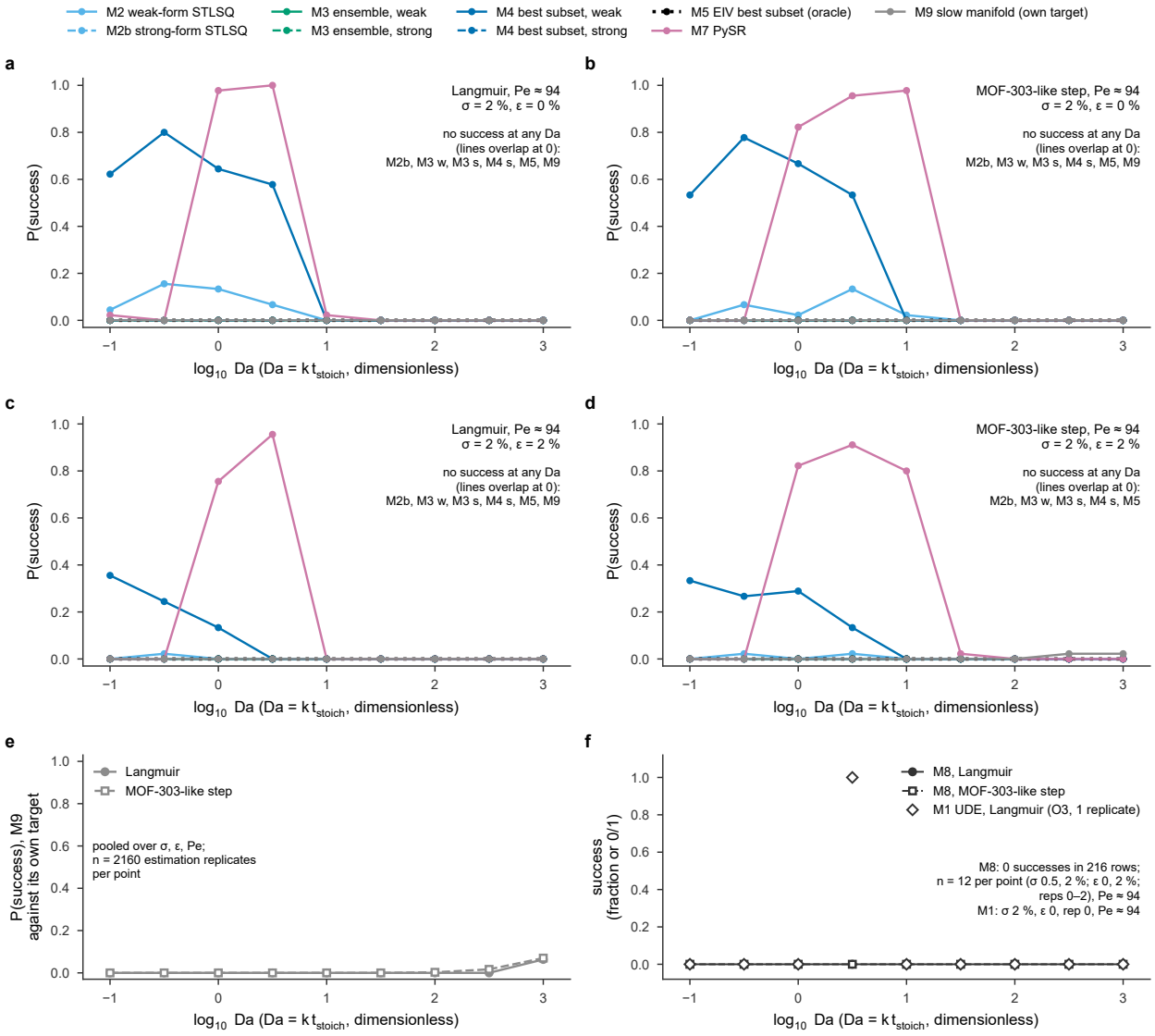
Extended Data Fig. 2: **Every H2b slice, MOF-303-like step isotherm.** As Extended Data Fig. 1.



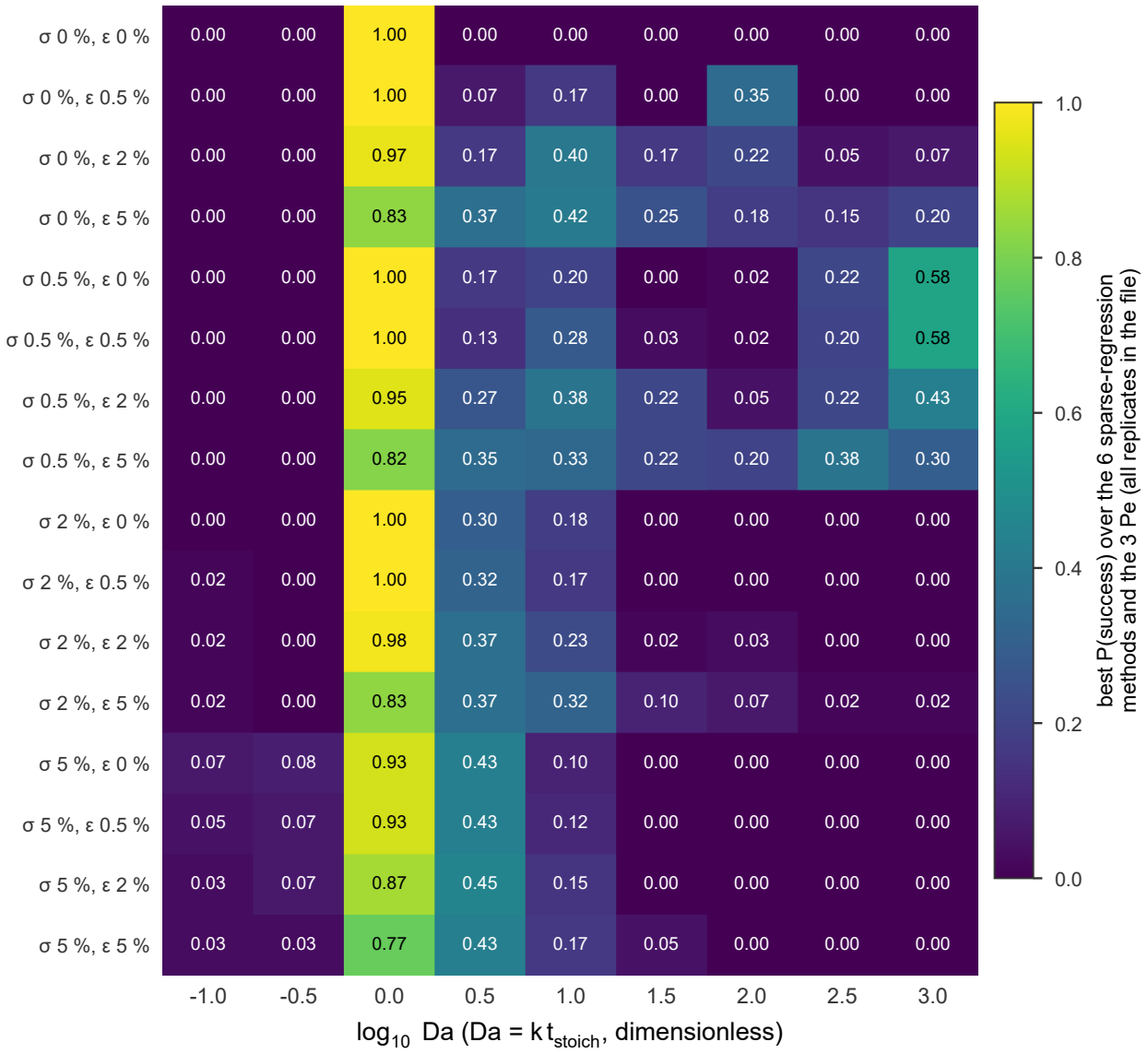
Extended Data Fig. 3: **Accuracy of the O1 identifiability estimator and H2b without the top two levels (descriptive checks made after the verdicts). a**, Median ratio of the estimated to the true k with exact data ($\sigma = \varepsilon = 0$), against Da, for both isotherms and three Péclet numbers; dark band, 0.9 to 1.1; light band, the criterion's $[0.5, 2]$. The estimate is accurate up to Da = 100 and biased low by the time resolution at 316 and 1000. **b**, For the 37 slices with a verdict, the H2b gap against the gap computed from the highest level up to Da = 100 at which k was identifiable in half the replicates (so without Da = 316 and 1000); every recomputed gap lies below the −0.25 decade threshold.



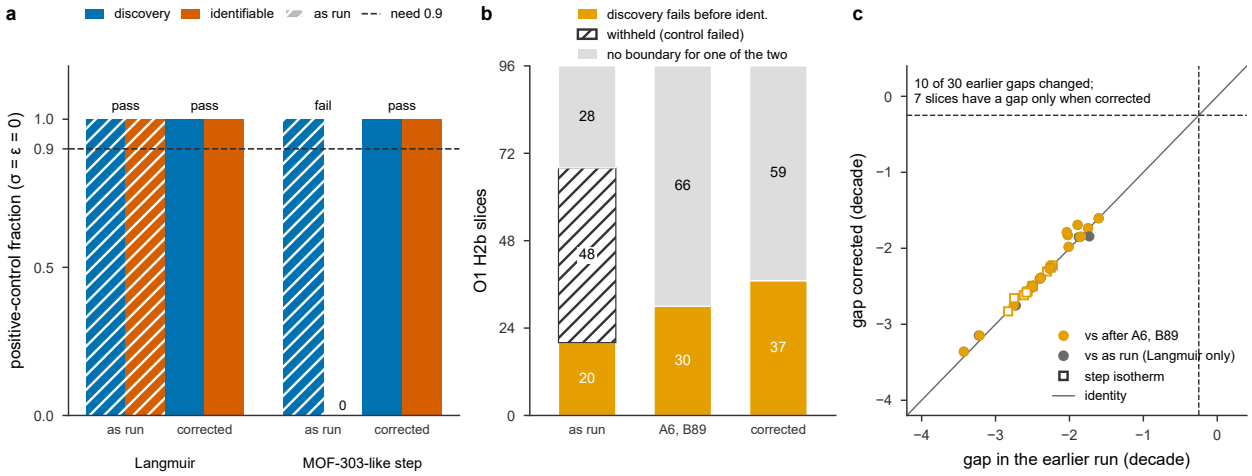
Extended Data Fig. 4: **Precision against coverage of the identifiability criterion (E3, exploratory).** Fraction of identifiability rows that meet the criterion (95 % interval inside $[k/2, 2k]$) and fraction whose interval contains the true k, by Da on O1 and in total on O2 and O3. Most intervals that meet the criterion do not contain k.



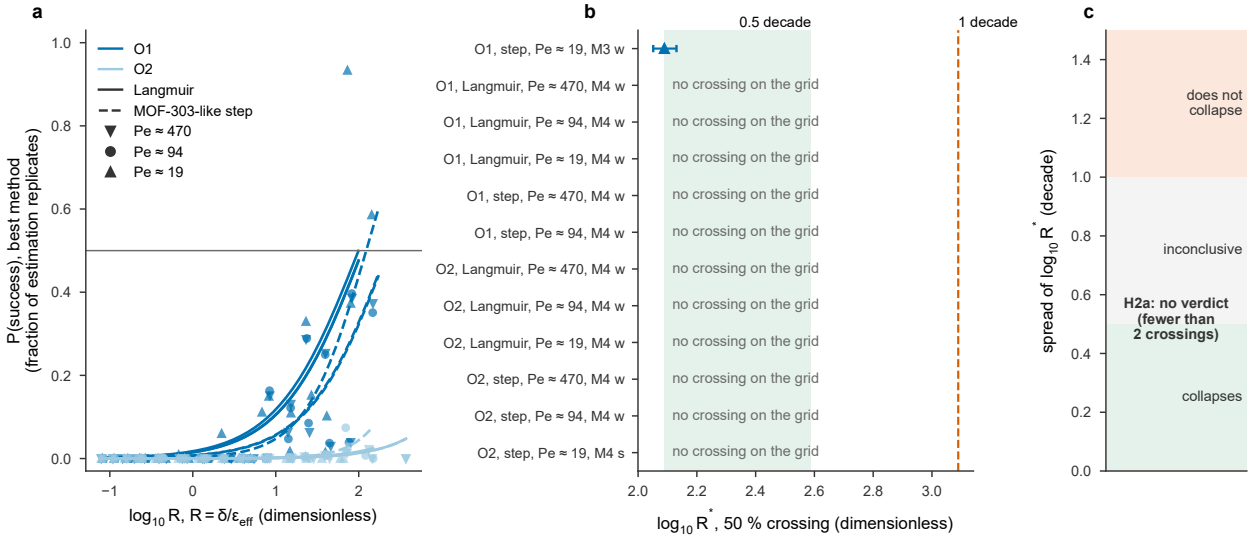
Extended Data Fig. 5: **Every method against** Da**, and the methods outside the confirmatory pool. a**–**d**, Success of each method at Pe ≈ 94 and 2 % noise, with an exact isotherm (**a**, Langmuir; **b**, step) and with 2 % isotherm error (**c**, **d**); methods listed in a panel had no success at any Da. **e**, Slow-manifold discovery (M9) against its own target, pooled over noise, isotherm error and Péclet number. **f**, The Kolmogorov–Arnold network (M8, exploratory E2; 0 of 216 fits) and the neural-network demonstration (M1; one replicate per level).



Extended Data Fig. 6: **Loading-dependent rate (H2d, exploratory).** Best success over methods and Péclet numbers for each noise and isotherm-error pair (rows) and Da (columns), over all 60 replicates; values printed in the cells.



Extended Data Fig. 7: **As-run against corrected results.** **a**, Positive-control fractions as first run (hatched) and corrected; the step isotherm failed as first run because of the identifiability search range (Supplementary Note 2). **b**, H2b slices by verdict as first run, after corrections A6 and B89, and after B90 (corrected, the primary result). **c**, Corrected gap of each slice against its gap in each earlier run (orange, after A6 and B89; grey, as first run, Langmuir only, the step slices having been withheld). No correction changed a verdict word.



Extended Data Fig. 8: **No single error ratio places the discovery boundary (H2a).** **a**, Success of the best method in each condition against $R = \delta/\varepsilon_{\mathrm{eff}}$, the realised driving force over the combined observation and isotherm error, on probes with uptake (O1, dark blue) and without (O2, light blue). **b**, The 50 % crossing R^* of each condition with its 95 % interval; only one of the twelve conditions crosses. **c**, The pre-registered decision regions for the spread of $\log_{10} R^*$; with fewer than two crossings, H2a has no verdict.